Slide 1: Title Slide

Topic: AI-Based Intelligent Software Defect Prediction

 Explanation:

This slide introduces your project. You are proposing a system that uses Machine Learning (ML) to predict software defects before testing.

 Example: Instead of finding bugs after testing, your system predicts:

 "This module has 80% chance of bug"

Viva Questions: What is Software Defect Prediction?

 It is the process of identifying defect-prone modules before testing using historical data.

Slide 2: Abstract

 Explanation: This is a summary of your project:

Uses ML models like:

Random Forest

Decision Tree

Logistic Regression

Uses past data (defects, code complexity)

Improves testing efficiency by 40%

 Example:

If a login module had many bugs earlier → ML predicts it will have bugs again.

Viva Questions:

Q2: Why use Machine Learning here?

 To analyze patterns in past defects and predict future defects automatically.

Slide 3: Introduction

 Explanation:

Software defects reduce quality

Traditional testing is late & costly

ML helps in early prediction

 Example:

Traditional

Find bug after deployment

AI-based:

 Predict bug before testing starts

 Viva Questions:

Q3: Why early defect detection is important?

 It reduces cost, effort, and improves software quality.

 Slide 4: Literature Review

 Explanation:

You studied previous research:

Algorithms used:

Naïve Bayes

SVM

Decision Tree

Random Forest

Problems:

Imbalanced data

Low accuracy

No real-time usage

 Example:

If dataset has:

90% non-defective

10% defective

→ Model may become biased

 Viva Questions:

Q4: Which algorithm gives better accuracy?

 Random Forest usually performs better.

 Slide 5: Existing System

 Explanation:

Current system:

Manual testing

Detects bugs after development

Time-consuming

Depends on tester skill

 Example:

Tester checks all modules equally → wastes time

 Viva Questions:

Q5: What are limitations of existing system?

 Late detection, high cost, more manual effort, less efficiency.

 Slide 6: Proposed System

 Explanation:

Your system:

Uses ML models

Predicts defects early

Focus on high-risk modules

 Example:

System output:

 “Payment module = HIGH RISK”

 Viva Questions:

Q6: What is your main contribution?

 AI-based prediction system to reduce testing effort and cost.

 Slide 7: Hardware Requirements

 Explanation:

Minimum system needed:

i3 processor

4GB RAM (8GB better)

20GB storage

 Example:

Even a normal laptop can run this project.

 Viva Questions:

Q7: Why 8GB RAM recommended?

 For faster ML model training.

 Slide 8: Software Requirements

 Explanation:

Language: Python

Libraries:

Pandas → data handling

NumPy → numerical operations

Scikit-learn → ML models

Matplotlib → graphs

 Example:

Using Pandas to clean missing values.

 Viva Questions:

Q8: Why Python is used?

 Easy, powerful libraries, best for ML.

 Slide 9: Block Diagram

 Explanation:

Flow of system:

Data Collection

Data Preprocessing

Model Training

Prediction

Evaluation

 Example:

Raw data → Clean data → Train model → Predict defects

 Viva Questions:

Q9: What is preprocessing?

 Cleaning data (removing missing/incorrect values).

 Slide 10: Module Description

 Explanation:

1. Data Collection

Collect past project data

2. Data Preprocessing

Clean & normalize data

3. Model Training

Train ML algorithms

4. Prediction

Predict defect or not

5. Evaluation

Check accuracy, precision, recall

 Example:

Input: Code metrics

Output: Defective / Not Defective

 Viva Questions:

Q10: What is precision?

 Correct positive predictions.

Q11: What is recall?

 How many actual defects are identified.

 Slide 11: Screenshots

 Explanation:

This slide shows:

Output of your system

Graphs / predictions

 Example:

Graph showing:

 Accuracy vs Model

 Viva Questions:

Q12: What results did you get?

 High accuracy and reduced testing effort.

 Slide 12: Conclusion

 Explanation:

AI improves software quality

Reduces cost & effort

Helps focus on risky areas

 Example:

Instead of testing 100 modules → test only 30 risky ones

 Viva Questions:

Q13: What is the benefit of your system?

 Saves time, cost, and improves reliability.

 Slide 13: References

 Explanation:

Sources used for research.

 Viva Questions:

Q14: Why references are important?

 To support your research and show credibility.

 Slide 14: Future Work

 Explanation:

Future improvements:

Deep Learning models

Real-time prediction

CI/CD integration

Cloud deployment

 Example:

Predict defects during coding itself.

 Viva Questions:

Q15: What can be improved in future?

 Accuracy, real-time integration, scalability.

 Slide 15: Thank You

 Explanation:

Closing slide.

 FINAL IMPORTANT VIVA QUESTIONS

 Must Prepare:

What is Machine Learning?

 System that learns from data and makes predictions.

What is Random Forest?

 Ensemble model using multiple decision trees.

What is overfitting?

 Model performs well on training data but fails on new data.

What is SMOTE?

 Technique to balance imbalanced datasets.

Difference: Precision vs Recall

 Precision = Correct predicted defects

 Recall = Actual defects found

Why defect prediction is important?

 Saves cost, improves quality, faster development